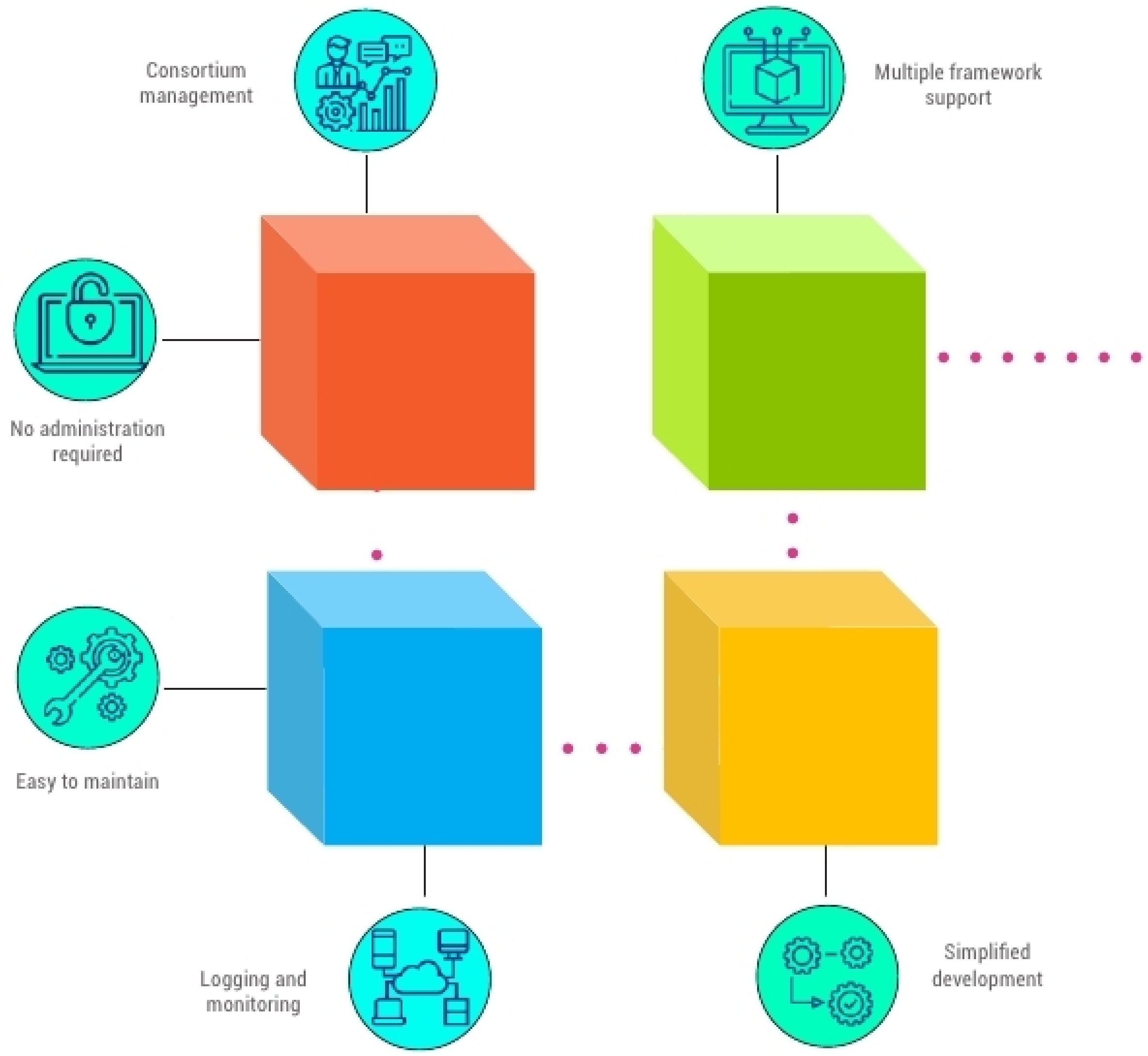
An Introduction to Microsoft's Azure Blockchain Service



Blockchain has captured global attention as a great way to streamline business processes, verify transactions and reduce any sort of fraud. This article gives a quick overview of Microsoft's Azure Blockchain Service.

Blockchain is a secure, shared, distributed ledger that can be public, private or consortium. It is highly secure as it uses cryptography to create transactions that are quite impossible to tamper with. All the information is shared among all the nodes or peers in the chain to form contracts. A blockchain is deeply linked to the number of entities that are participating in it. All the blockchain data and contracts are distributed, forming numerous replicas in the database. A greater number of replicas ensures the reliability of data. A blockchain can be defined as a 'digital ledger' — a transactional database that incorporates only immutable records of every transaction that occurs.

In a blockchain, ledgers are distributed across the entire network and no middleman is needed for the transaction. It acts like a peer-to-peer file sharing system, as every peer obtains a copy of the entire data set.

Blockchain technology evolved with the addition of smart contracts, which are small pieces of code that add logic to transactions. Smart contracts can be termed as computer code representations of legal terms in contract for goods or services. With the passage of time, various new blockchain ledgers have emerged in the market like Ethereum and Hyperledger Fabric known as Blockchain 2.0. But just as databases have evolved over time by adding a logic execution capability (in the form of stored procedures, for example), blockchain has introduced smart contracts to handle the logic tier. However, smart contracts can only operate on data contained in the block where they are stored. They can't access external data or systems, as calling a service outside of the blockchain breaks the 'circle of trust' that blockchain provides for cryptographic security and immutability of transactions. CRM, ERP and payroll systems all represent external entities that aren't part of a blockchain, but may be involved in the exchange of data within a transaction. Blockchains need a way to securely receive external data, as well as access to secure execution of off-chain code.